Project 2: Turbojet Engine Design/Off-Design Performance Analysis

Riley J. Kyle*
Missouri University of Science and Technology, Rolla, Missouri, 65409, United States

Dr. David W. Riggins†
Missouri University of Science and Technology, Rolla, Missouri, 65409, United States

The purpose of this project is to plot the performance three space of a fixed-area missile-class engine with a simple converging nozzle. This will be done by varying the fuel flow rate from 20% to 500% the on design fuel rate, varying the Mach number from 0 to 2.5. Additionally, this will be done at sea level, 4000 m, 8000 m, and 12000 m altitude.

Nomenclature

T = temperature $[K]$

P = pressure $[Pa]$

ρ = density $\left[kg/m^3\right]$

T_t = total temperature $[K]$

P_t = total pressure $[Pa]$

τ = $\frac{T_{t,out}}{T_{t,in}}$

π = $\frac{P_{t,out}}{P_{t,in}}$

η = isentropic efficiency

u = velocity $[m/s]$

M = mach number

Subscripts

0 = freestream

d = diffuser

c = compressor

b = burner

t = turbine

n = nozzle

*Aerospace Engineering Student, MAE
†Professor of Mechanical and Aerospace Engineering, MAE

I. Introduction

The purpose of this project is to plot the performance three space of a fixed-area missile-class engine with a simple converging nozzle. This will be done by varying the fuel flow rate from 20% to 500% the on design fuel rate, varying the Mach number from 0 to 2.5. Additionally, this will be done at sea level, 4000 m, 8000 m, and 12000 m altitude.

II. Completely define the on-design engine fluid dynamics and performance

A. Methodology

B. Results

III. Develop the complete performance envelope for this specific turbojet engine

A. Methodology

B. Results

1. Sea Level

2. 4000 m

3. 8000 m

4. 12000 m

IV. Conclusion

Appendix